

CHE 201 Spring 2020 HW 5

Roxanna Mendoza

TOTAL POINTS

95 / 100

QUESTION 1

1 Question 1 15 / 15

- ✓ - **0 pts** Correct
- **5 pts** wrong specific volume
- **5 pts** wrong T2
- **5 pts** wrong T when the tank contains only saturated vapor
- **15 pts** Question not answered

QUESTION 2

2 Question 2 15 / 15

- ✓ - **0 pts** Correct
- **5 pts** wrong v1
- **5 pts** wrong v2
- **5 pts** wrong work
- **15 pts** Question not answered

QUESTION 3

3 Question 3 15 / 15

- ✓ - **0 pts** Correct
- **5 pts** Wrong T-v diagram
- **2.5 pts** wrong U1
- **2.5 pts** wrong u2
- **2.5 pts** wrong electricity work
- **2.5 pts** wrong time
- **15 pts** Question not answered

QUESTION 4

4 Question 4 20 / 20

- ✓ - **0 pts** Correct
- **20 pts** Question not answered
- **5 pts** wrong u1
- **5 pts** wrong u2
- **5 pts** wrong electricity work
- **5 pts** wrong final heat transfer

QUESTION 5

5 Question 5 15 / 20

- **0 pts** Correct
- **5 pts** wrong sketch of p-v diagram
- ✓ - **2.5 pts** wrong Q1
- **2.5 pts** wrong W1
- **2.5 pts** wrong Q2
- ✓ - **2.5 pts** wrong W2
- **2.5 pts** wrong Q3
- **2.5 pts** wrong W3
- **20 pts** Question not answered

QUESTION 6

6 Question 6 15 / 15

- ✓ - **0 pts** Correct
- **2.5 pts** not enough components of ways to improve water use
- **5 pts** not enough components of ways to improve water use
- **7.5 pts** not enough components of ways to improve water use
- **2.5 pts** not enough components of record of water use and compare
- **5 pts** not enough components of record of water use and compare
- **15 pts** Question not answered

Homework #9

Roxanna Mendoza

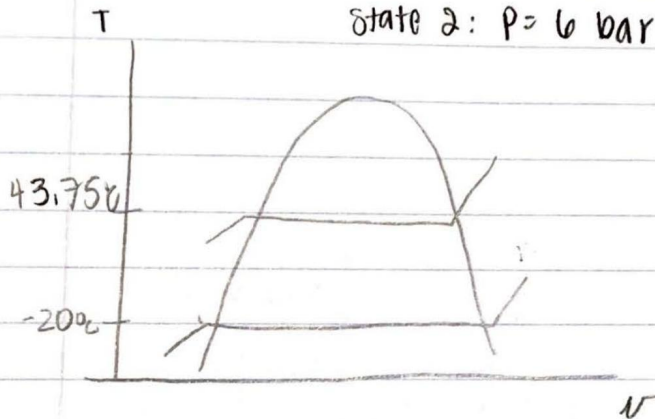
#1. Refrigerant 22 CHClF2

Sketch:



state 1: $T = -20^\circ\text{C}$
 $x = 50.36\%$

state 2: $P = 6 \text{ bar}$



Engineering model

- ① closed system
- ② rigid tank means constant volume
- ③ Liquid-vapor mixture
- ④ substance modeled as incompressible,
 $C_p = C_v \leftarrow$ common specific heat

Table A-7

@ $-20^\circ\text{C} \Rightarrow 2.4534 \text{ Bar}$
 $V_f = 0.7427 \times 10^{-3} \text{ m}^3/\text{kg}$
 $V_g = 0.0926 \text{ m}^3/\text{kg}$
 $x = 50.36\% \Rightarrow x = .5036$

$$V_1 = V_f + x_1 (V_g - V_f)$$

$$V_1 = (0.7427 \times 10^{-3}) + (0.5036)(0.0926 - 0.7427 \times 10^{-3})$$

$$V_1 = 0.047 \text{ m}^3/\text{kg}$$

@ $P = 6 \text{ bar}$ $V_f = 0.7927 \times 10^{-3} \text{ m}^3/\text{kg}$ $V_g = 0.0392 \text{ m}^3/\text{kg}$
 $V_2 = V_1 = 0.047 \text{ m}^3/\text{kg}$

Looking at Table A-9 for $P = 6 \text{ bar}$ & $V = 0.047 \text{ m}^3/\text{kg}$
 Temperature $\times 40^\circ\text{C} \Rightarrow 0.04629 \text{ m}^3/\text{kg}$
 $\times 45^\circ\text{C} \Rightarrow 0.04724 \text{ m}^3/\text{kg}$

Assuming linear interpolation:

$$T_2 = T_x - (T_x - T_y) \cdot \frac{V_x - V_2}{V_x - V_y}$$

$$T_2 = (40) - (40 - 45) \cdot \left(\frac{0.04629 - 0.047}{0.04629 - 0.04724} \right) = 43.75^\circ\text{C}$$

Part B: $V_g = V_1 = 0.047 \text{ m}^3/\text{kg}$
 $(T_s = 0^\circ\text{C})$ $P_s = 4.9911 \text{ bar}$

1 Question 1 15 / 15

✓ - 0 pts Correct

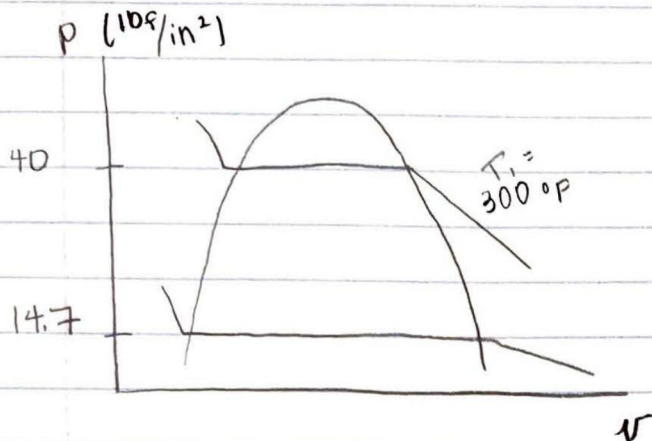
- 5 pts wrong specific volume
- 5 pts wrong T2
- 5 pts wrong T when the tank contains only saturated vapor
- 15 pts Question not answered

#2.



Engineering Model

- ① The water is a closed system
- ② Polytropic process $pV^{1.2} = \text{constant}$



$$P_1 = 40 \text{ lbf/in}^2$$

$$P_2 = 14.7 \text{ lbf/in}^2$$

$$T_1 = 300^\circ\text{F}$$

$$T_2 = ??$$

From Table A-4E

$$v_1 = 11.04 \text{ ft}^3/\text{lbm}$$

so

$$P_1 v_1^{1.2} = P_2 v_2^{1.2}$$

$$\Rightarrow 14.7 v_2^{1.2} = 40 (11.04)^{1.2}$$

$$v_2^{1.2} = \frac{40 (11.04)^{1.2}}{14.7}$$

$$v_2 = 25.425 \text{ ft}^3/\text{lbm}$$

$$P_1 = 40 \text{ lbf/in}^2 = \frac{40 \text{ lbf}}{\text{in}^2} \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)^2 = 5760 \text{ lbf/ft}^2$$

$$P_2 = 14.7 \text{ lbf/in}^2 = \frac{14.7 \text{ lbf}}{\text{in}^2} \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)^2 = 2116.8 \text{ lbf/ft}^2$$

$$W = \left[\frac{P_1 v_1 - P_2 v_2}{n-1} \right] (m)$$

$$W = \left[\frac{(5760 \text{ lbf/ft}^2)(11.04 \text{ ft}^3/\text{lbm}) - (2116.8 \text{ lbf/ft}^2)(25.425 \text{ ft}^3/\text{lbm})}{1.2-1} \right] (0.5)$$

$$W = 24429 \text{ ft} \cdot \text{lbf} \Rightarrow 31.393 \text{ Btu}$$

2 Question 2 15 / 15

✓ - **0 pts** Correct

- **5 pts** wrong v1

- **5 pts** wrong v2

- **5 pts** wrong work

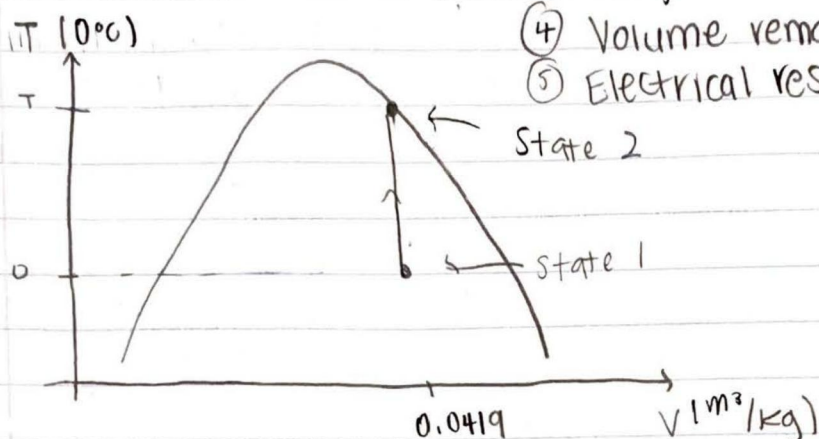
- **15 pts** Question not answered

#3.



Engineering model

- ① closed system
- ② liquid-vapor mixture
- ③ Refrigerant 134a
- ④ Volume remains constant
- ⑤ Electrical resistor

From Table $T = 0^\circ\text{C}$ & $x = 0.10$

$$u_1 = u_f + x(u_{fg}) \quad \leftarrow @ 0^\circ\text{C}$$

$$u_1 = 159 \text{ kJ/kg}$$

$$v_1 = v_f + x(v_{fg}) \quad \leftarrow @ 0^\circ\text{C}$$

$$v_1 = 0.0419 \text{ m}^3/\text{kg}$$

$$v_2 = 0.0419 \text{ m}^3/\text{kg} \quad @ \quad x_2 = 1$$

$$u_2 = 238 \text{ kJ/kg}$$

Internal energy is increased by electric resistor

$$\Delta U = m(u_2 - u_1)$$

$$= (1.5)(238 - 159) = 118.5 \text{ kJ}$$

$$\text{Power} = 2 \text{ kW}$$

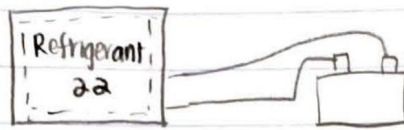
$$\text{time} = \frac{118.5 \text{ kJ}}{2 \text{ kW}} = 59.25 \text{ s}$$

3 Question 3 15 / 15

✓ - 0 pts Correct

- 5 pts Wrong T-v diagram
- 2.5 pts wrong U_1
- 2.5 pts wrong u_2
- 2.5 pts wrong electricity work
- 2.5 pts wrong time
- 15 pts Question not answered

4.



Engineering model

- ① closed system
- ② constant volume

Refrigerant

Initial Temperature $T_1 = -10^\circ\text{C}$

Quality $x = 0.9$

Volume $V = 0.01 \text{ m}^3$

Final temperature $T_2 = 40^\circ\text{C}$

Battery

Voltage: 12 V

Current: 5 A

Time $t = 5 \text{ min}$

$$U_1 = U_f + x(U_g - U_f) \Rightarrow U_f = 33.27 \text{ kJ/kg} \quad \& \quad U_g = 223.02 \text{ kJ/kg}$$

$$U_1 = 33.27 + 0.9(223.02 - 33.27) = 195.07 \text{ kJ/kg} \Rightarrow U_1$$

$$V_1 = V_f + x(V_g - V_f) \Rightarrow V_f = 0.7606 \times 10^{-3} \text{ m}^3/\text{kg} \quad \& \quad V_g = 0.0652 \text{ m}^3/\text{kg}$$

$$V_1 = (0.7606 \times 10^{-3}) + 0.9(0.0652 - 0.7606 \times 10^{-3}) = 0.0523 \text{ m}^3/\text{kg} \Rightarrow V_1$$

$$V_2 = V_1$$

$$= 0.052312 \text{ m}^3/\text{kg}$$

$$\text{c) } T_2 = 40^\circ\text{C}$$

$$V_2 = 0.052312 \text{ m}^3/\text{kg}$$

$$U_2 = 250.390 \text{ kJ/kg}$$

$$\text{mass} = \frac{\text{Volume}}{\text{Specific Volume}}$$

$$m = \frac{0.01 \text{ m}^3}{0.0523 \text{ m}^3/\text{kg}} = 0.1912 \text{ kg}$$

$$\begin{aligned} W &= -V_1 \Delta t \\ &= -(12 \text{ V} \times 5 \text{ A} \times 5 \text{ min}) \times \frac{1 \text{ W/A}}{1 \text{ V}} \times \frac{60 \text{ sec}}{1 \text{ min}} \times \frac{1 \text{ J/s}}{1 \text{ W}} \times \frac{1 \text{ kJ}}{10^3 \text{ J}} \\ &= -18 \text{ kJ} \end{aligned}$$

$$Q = \Delta U_{12} + W_{12}$$

$$= m(U_2 - U_1) + W$$

$$= 0.1912 \text{ kg} (250.39 - 195.07) \text{ kJ/kg} - 18 \text{ kJ}$$

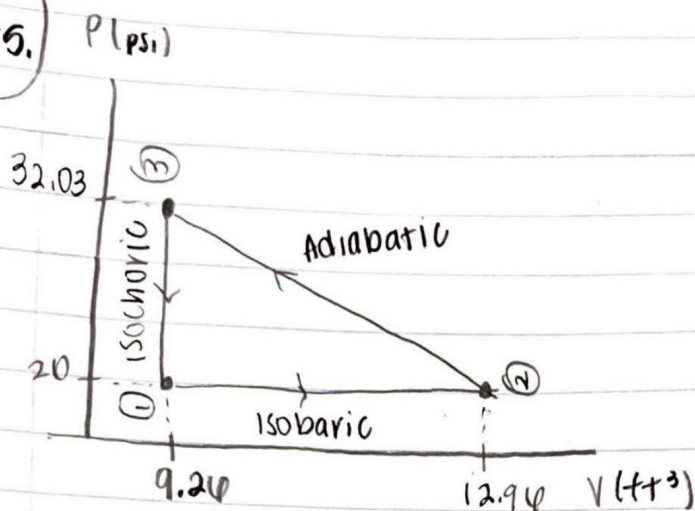
$$(Q = -5.51 \text{ kJ})$$

4 Question 4 20 / 20

✓ - 0 pts Correct

- 20 pts Question not answered
- 5 pts wrong u1
- 5 pts wrong u2
- 5 pts wrong electricity work
- 5 pts wrong final heat transfer

#5.



Process 1 $\rightarrow$ 2 Isobaric Process $\therefore$ constant Pressure

$$T_1 = 500^\circ\text{R} = 40.33^\circ\text{F}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow T_2 = T_1 \frac{V_2}{V_1}$$

$$T_2 = (40.33^\circ\text{F}) \left(\frac{12.96}{9.26} \right) = 56.46^\circ\text{F}$$

$$Q = m C_p (T_2 - T_1)$$

$$= 110 \times 0.2400 \text{ BTU/lbm}^\circ\text{F} (56.46 - 40.33)$$

$$Q = 3.87 \text{ BTU}$$

Find V_1 using Ideal gas equation

$$V_1 = \frac{m R T_1}{P_1}$$

$$= 110 \times (0.031081 \text{ slug/lb}) \left(\frac{1716 \text{ BTU/lbm}^\circ\text{R} \times 500 \text{ R}}{20 \text{ lbf/in}^2 \times 144 \text{ in}^2/\text{ft}^2} \right)$$

$$V_1 = 9.26 \text{ ft}^3$$

Final volume

$$V_2 = \left(\frac{12.96}{9.26} \right) (9.26 \text{ ft}^3) = 12.96 \text{ ft}^3$$

$$W = p dV$$

$$= p_1 (V_2 - V_1) = 2000 (12.96 - 9.26) = 10656 \text{ lb-ft}$$

$$= 13.694 \text{ BTU}$$

Process 2 $\rightarrow$ 3 Adiabatic process

$$Q = 0$$

$$dU + dW = dQ$$

$$dW = -dU$$

$$= -m C_v (T_3 - T_2)$$

$$= -[110 \times 0.1715 \text{ Btu/lbm}^\circ\text{F} (360.33 - 56.46^\circ\text{F})]$$

$$W = -52.113 \text{ Btu}$$

Process 3 $\rightarrow$ 1 Isochoric process

$$W = 0$$

$$dU + dW = dQ$$

$$dQ = dU$$

$$= m C_v (T_1 - T_3)$$

$$= -110 \times 0.1715 \text{ Btu/lbm}^\circ\text{F} (40.33 - 360.33)$$

$$Q = -54.88 \text{ Btu}$$

5 Question 5 15 / 20

- 0 pts Correct
- 5 pts wrong sketch of p-v diagram
- ✓ - 2.5 pts wrong Q1
 - 2.5 pts wrong W1
 - 2.5 pts wrong Q2
- ✓ - 2.5 pts wrong W2
 - 2.5 pts wrong Q3
 - 2.5 pts wrong W3
- 20 pts Question not answered

Roxanna Mendoza

February 23, 2020

CHE 201

Homework #5 Problem #6 3.9D

Water use is one of the main concerns throughout the world. There are many countries that do not have the luxury to be able to use as much water as they want but in other countries where the amount of water use is not restricted it is taken advantage of. As a society we must seek ways to reduce our water footprint. Starting off with reducing water use in the household and then going bigger and finding ways to use water efficiently in businesses and industries.

Water is poorly managed in households. Some examples of this would be letting your shower run till one finds the temperature at which they like taking their showers also letting the water run while one takes a shower. A way to reduce water use in this situation would be to turn the water off while washing your hair and body as well as to take a shorter amount of time in the shower. For leaking faucets replacing the faucet will reduce the amount of water that is not used. Only running dishwashers and laundry machines when there are full loads of dishes/clothing also using the lowest appropriate water level in these machines. As for outdoors not having the sprinklers on all the time helps reduce the amount of water used a tip as stated in the article would be to water the lawn when it's cooler out throughout the day because that is when more water is absorbed by the roots.

The water usage in businesses vary tremendously depending on the type of business. Some ways in which we could lower the amount of water used would be to install water saving toilets as incorporated in many schools where the amount of water dispensed depends on the type

of waste being flushed down. Another suggestion would be for businesses like car washes to retain the water used and reuse it when it is the first rinse of the car. Also instead of rinsing out sidewalks and driveways one could sweep all the waste.

Lastly, industries have machinery that requires a good amount of water. Although this is the case there are ways to save water which would include modifying equipment for low water consumption. Industries often need water for heat transfer, power generations, and industrial processes which vary. Educating one's employees on water conservation helps reduce water consumption because they are contributing to effective ways in which they can help.

Water consumption is an issue our society has been trying to fix and one can start off with making small changes within the household. From there on it could build on to businesses and industries. While keeping track of water consumption I made for 3 days it was about 100 gallons which is way more than the one gallon many people are limited to in other countries.

References

Matani, A G. *Improving Water Use Efficiency in Industries*.

Quinn, Tracy. "How Can You Reduce Your Water Footprint?" *NRDC*, 15 Dec. 2016,
www.nrdc.org/experts/tracy-quinn/how-can-you-reduce-your-water-footprint.

Sheldon, Elizabeth. "Ways That Communities or the Government Can Conserve Water."
Sciencing, 2 Mar. 2019,
sciencing.com/ways-communities-government-can-serve-water-19029.html.

"Water Conservation Tips for Businesses." *EPA*, Environmental Protection Agency, 10 Apr.
2017, www3.epa.gov/region1/eco/drinkwater/water_conservation_biz.html.

6 Question 6 15 / 15

✓ - 0 pts Correct

- 2.5 pts not enough components of ways to improve water use
- 5 pts not enough components of ways to improve water use
- 7.5 pts not enough components of ways to improve water use
- 2.5 pts not enough components of record of water use and compare
- 5 pts not enough components of record of water use and compare
- 15 pts Question not answered